# HTTP Protocol
# (HyperText Transfer Protocol)

(Module 1.b)







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# Objective

## By the end of this class, students will be able to

- **Define** the HTTP protocol and explain its purpose.
- **Explain** how HTTP works using the client–server request–response model.
- **Identify** common HTTP methods and status codes.
- **Analyze** HTTP communication using a practical demonstration with browser developer tools or Wireshark.

# Why Do We Need HTTP?

HTTP is needed because it provides a standard way for:

- Web browsers to request information.
- Web servers to send information.
- Users to access web pages and online services.
- Applications to exchange data over the Internet.

# HTTP (HyperText Transfer Protocol)

HTTP is a protocol that allows web browsers and web servers to communicate and exchange information.

# HTTP (HyperText Transfer Protocol)

- HTTP is a communication protocol used on the **World Wide Web**.
- It enables communication between a **client** and a **web server**.
- A web browser sends a request to the server.
- The server sends a response back to the browser.

# How HTTP Works



User



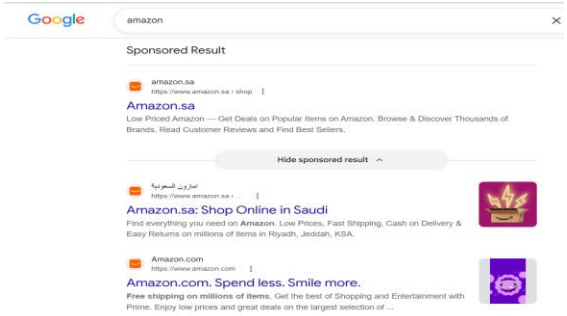
Browser




HTTP Request



Web Server




HTTP Response



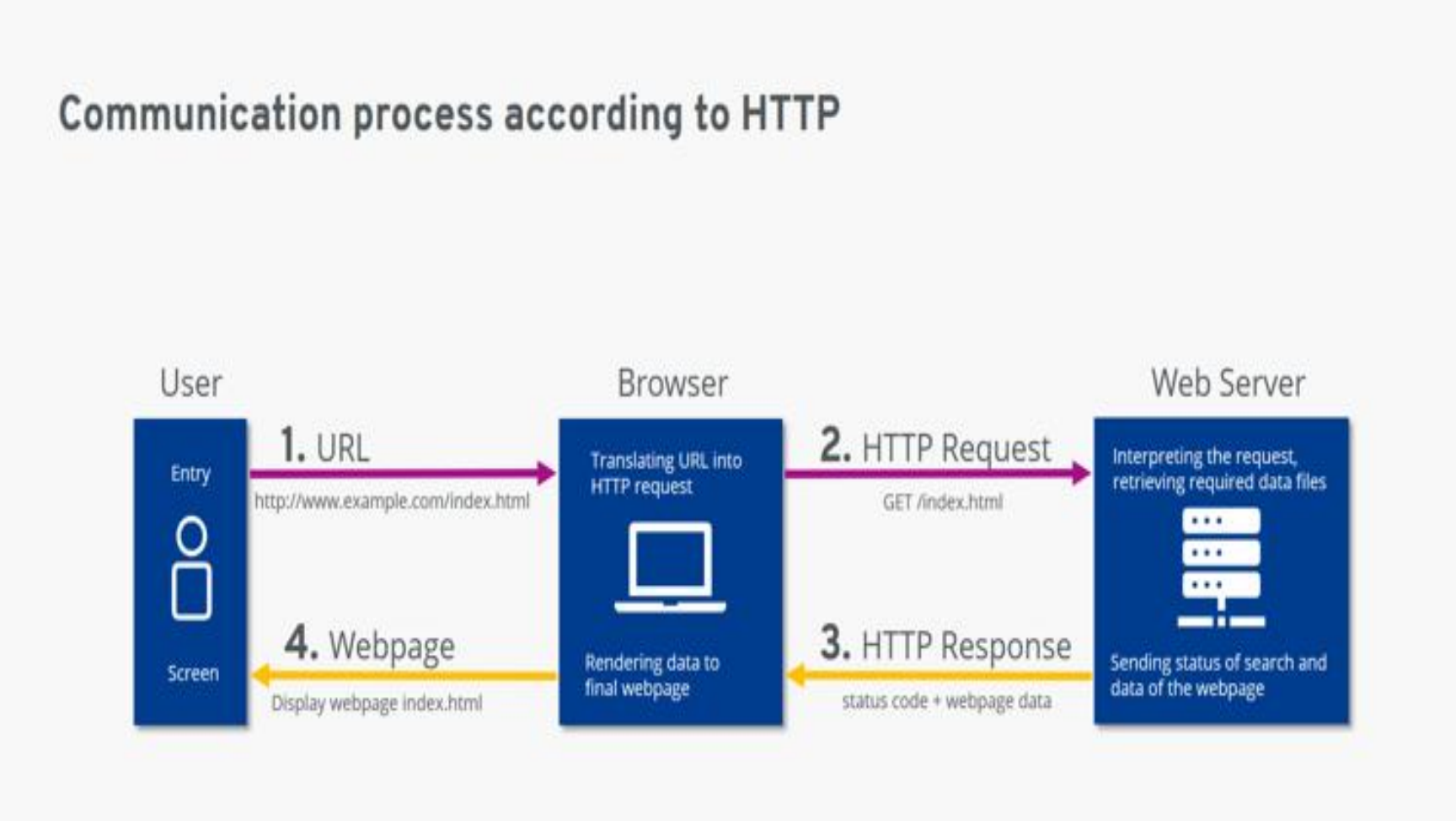
User

# How HTTP Works

## HTTP Communication Process

1. User enters a website URL.
2. The browser creates an HTTP request.
3. The request is sent to the web server.
4. The server processes the request.
5. The server sends an HTTP response.
6. The browser displays the requested webpage

# How HTTP Works



# HTTP – Video Demo

HTTP and HTTPS Explained in 2 Minutes

# HTTP Request

An HTTP request is a message sent by a client to request a resource from a web server.

**Example**

```
GET /index.html HTTP/1.1
Host: example.com
```

**Important Parts**

- **GET – HTTP Method**
- **/index.html – Requested resource**
- **HTTP/1.1 – HTTP version**
- **Host – Server name**

# HTTP Methods

## Common HTTP Methods

| Method | Purpose |
|---|---|
| GET | Retrieve information |
| POST | Send data to the server |
| PUT | Update existing data |
| DELETE | Remove data |
| HEAD | Retrieve headers only |

# HTTP Methods

## Common HTTP Methods

## Example

- **GET** → View a webpage
- **POST** → Submit a login form
- **PUT** → Update user information
- **DELETE** → Delete information

# HTTP Response and Status Codes

## HTTP Response

After receiving a request, the server sends a response.

## Example

**HTTP/1.1 200 OK**

**Content-Type: text/html**

**1. HTTP/1.1**

Indicates the **HTTP protocol version** used for communication.

Here, the server is using **HTTP version 1.1**.

**2. 200 OK**

- **200** is the HTTP **status code**.
- **OK** means the request was successfully processed by the server.

**3. Content-Type: text/html**

- Indicates the type of content included in the response.
- **text/html** means the server is sending an **HTML webpage** to the browser.

# HTTP Response and Status Codes

## HTTP Response

| Status Code | Meaning |
|---|---|
| 200 | OK – Request successful |
| 404 | Page Not Found |
| 403 | Access Forbidden |
| 500 | Internal Server Error |

# HTTP Characteristics

## Features of HTTP

- **Client–Server Protocol**
- **Stateless Protocol**
- Uses a **Request–Response model**
- Supports different types of data
- Simple and widely used
- Default HTTP port is **80**

## Important

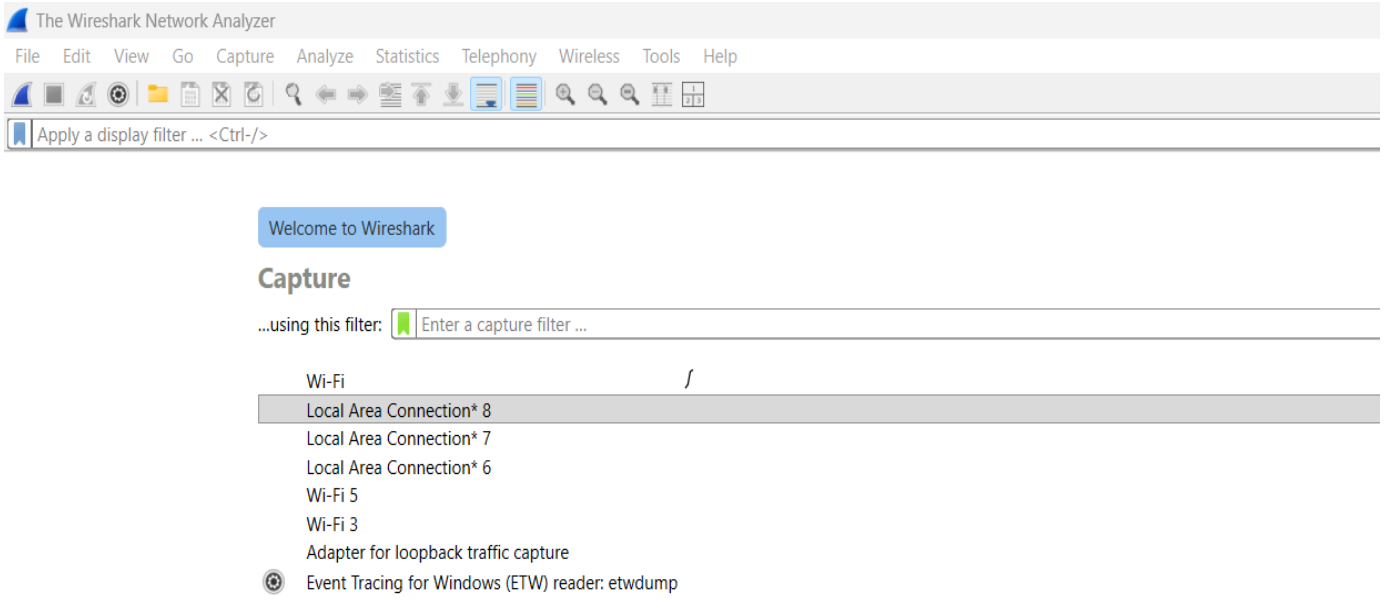
- **HTTP does not provide encryption by default.**
- For secure communication, **HTTPS** is used.

# HTTP Demonstration Using Wireshark

## Capturing HTTP Traffic

### Step 1:

Install and open **Wireshark**.



# HTTP Demonstration Using Wireshark

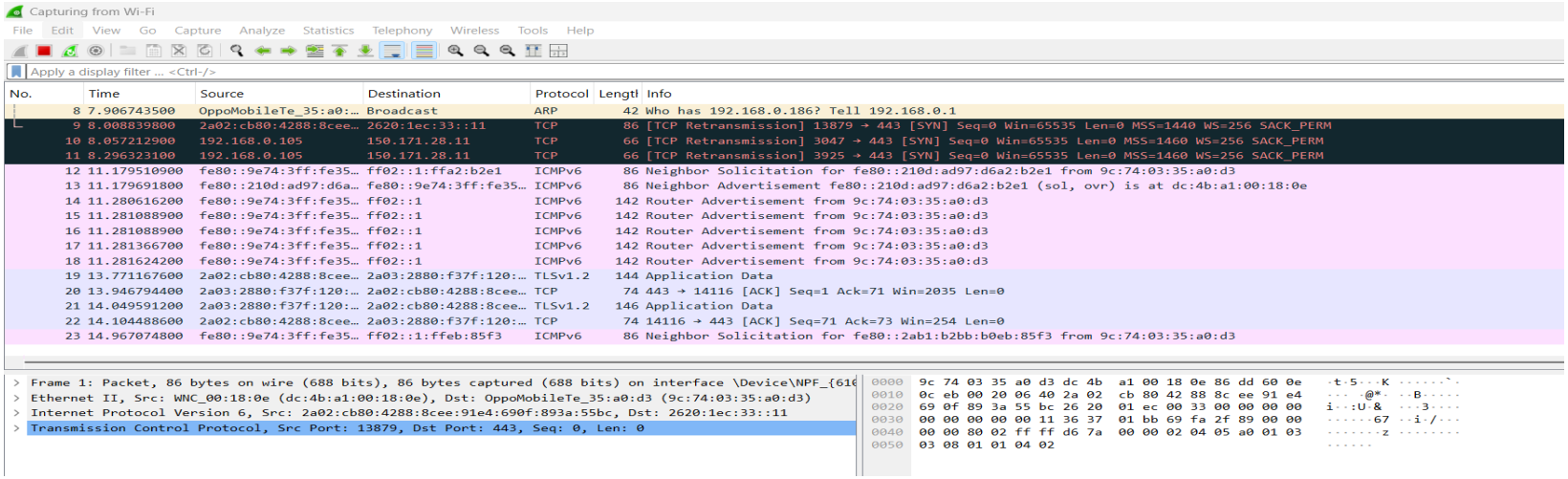
## Capturing HTTP Traffic

### Step 2:

Select the active network interface.

### Step 3:

Start capturing packets.



# HTTP Demonstration Using Wireshark

## Capturing HTTP Traffic

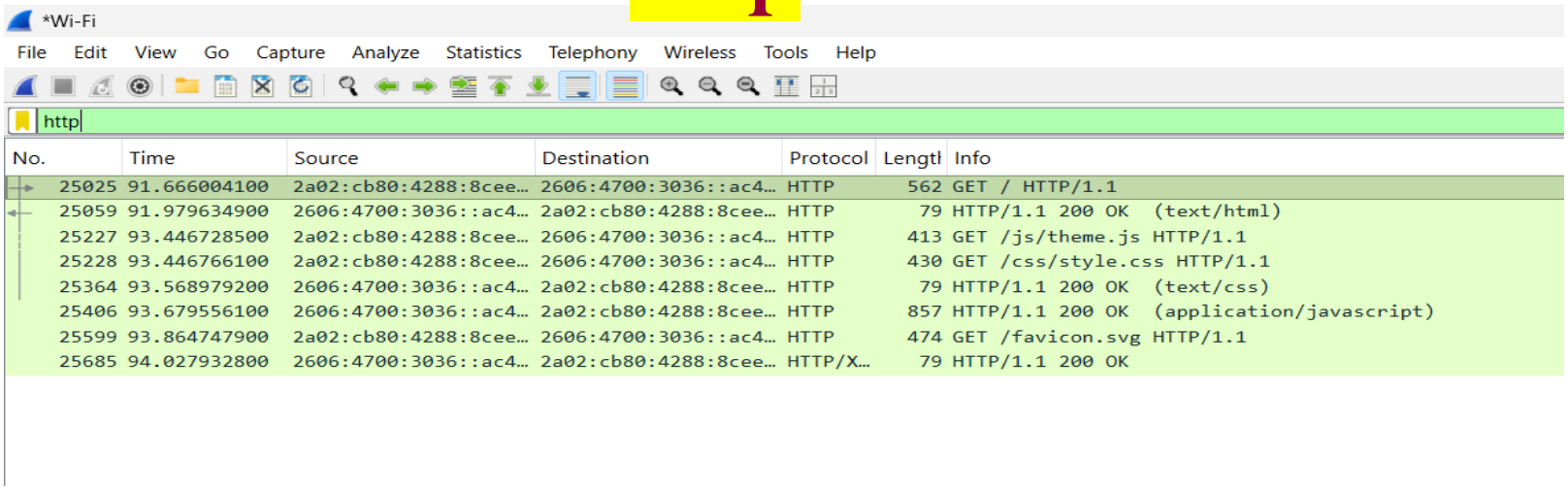
**Step 4:**

Open a browser and access an HTTP website.

**Step 5:**

Apply the filter:

**http**

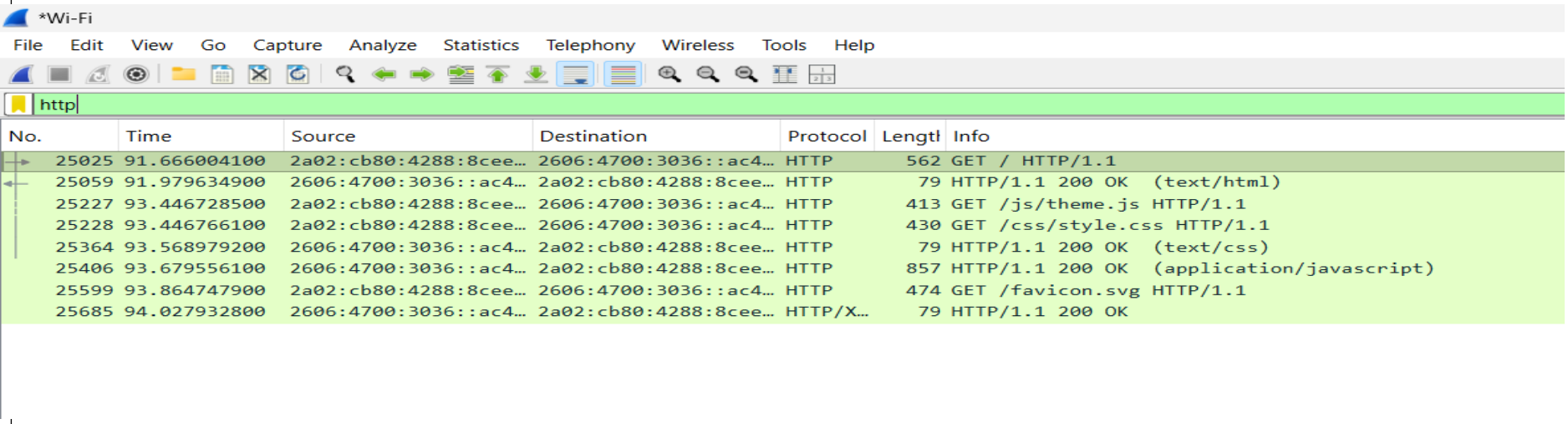


# HTTP Demonstration Using Wireshark

## Step 6:

Observe HTTP packets such as:

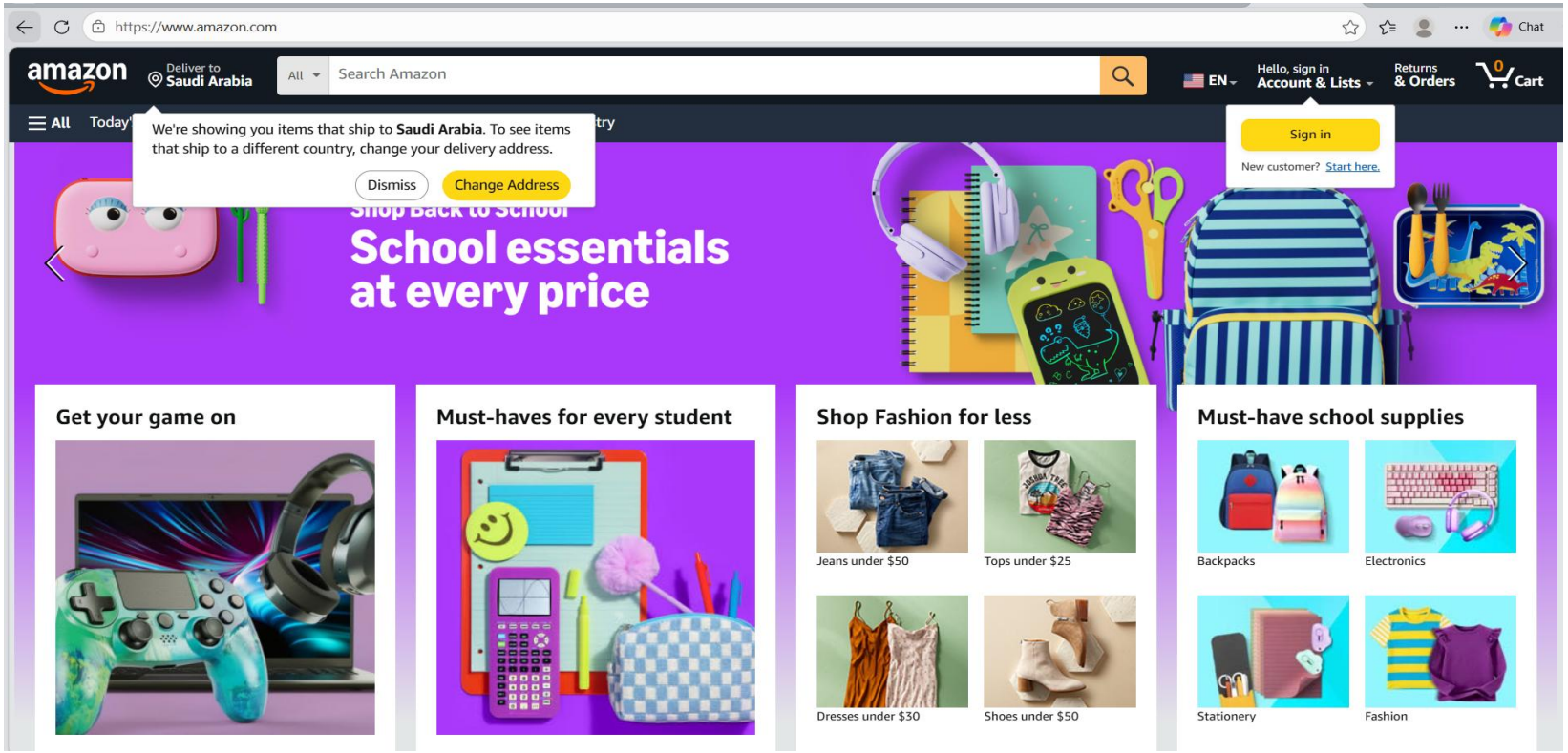
- HTTP GET Request
- HTTP Response
- Host information
- Requested webpage



# HTTP Protocol Demo Using Browser Developer Tools

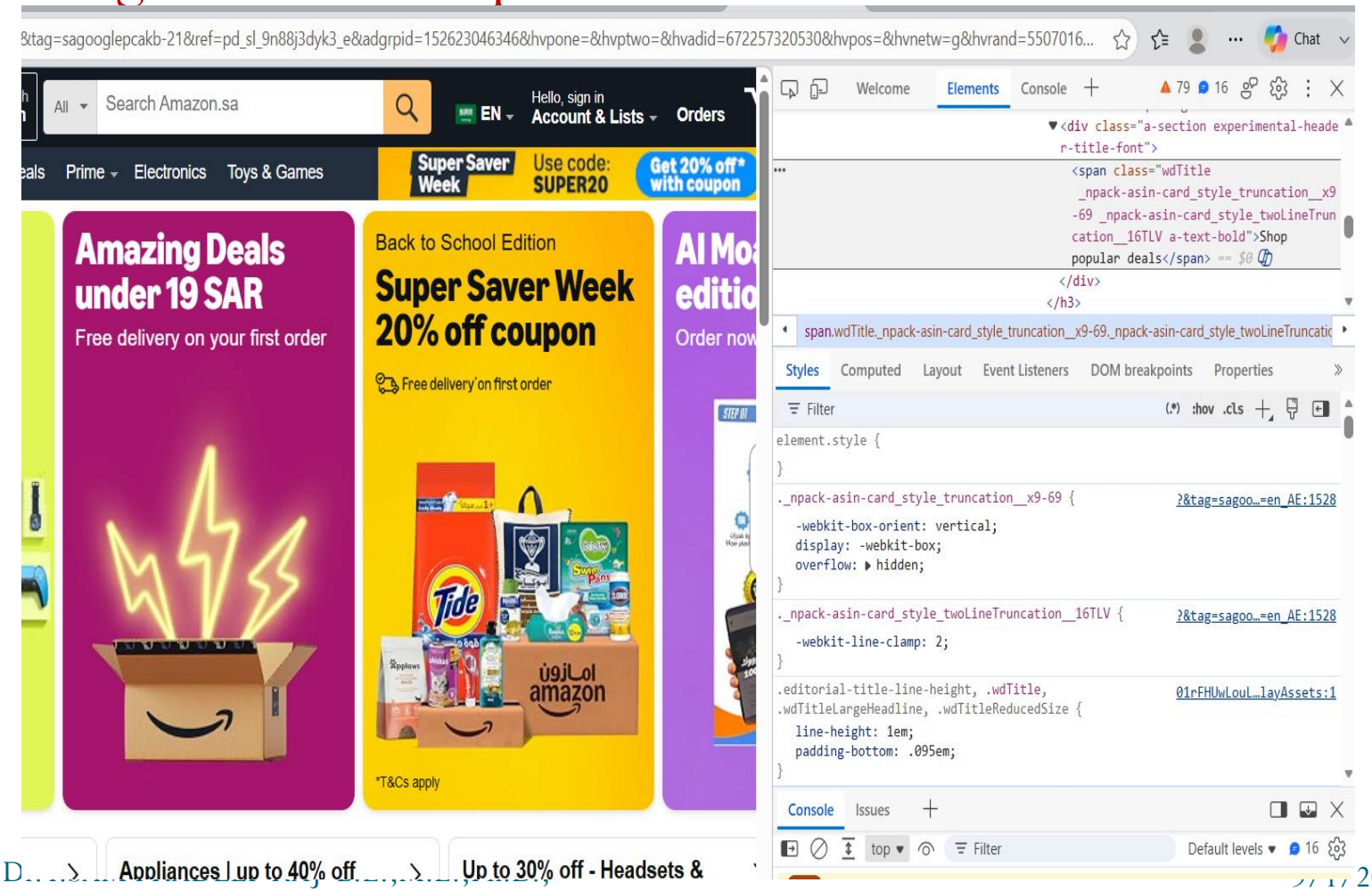
**Step 1:** Open Google Chrome or Microsoft Edge.

**Step 2:** Open any website.



# HTTP Protocol Demo Using Browser Developer Tools

**Step 3:** Right-click → Inspect.

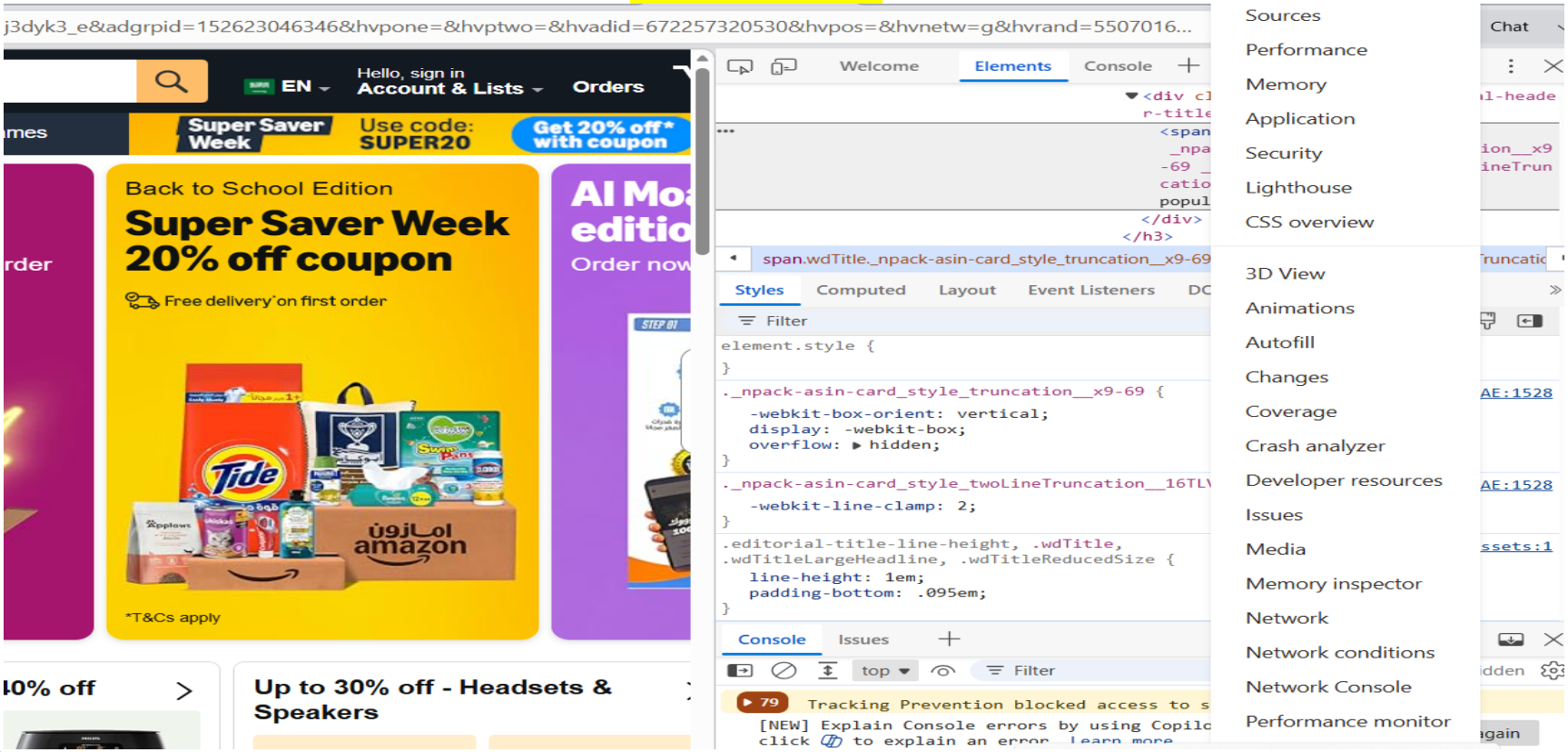


# HTTP Protocol Demo Using Browser Developer Tools

## Step 4: Click the Network tab

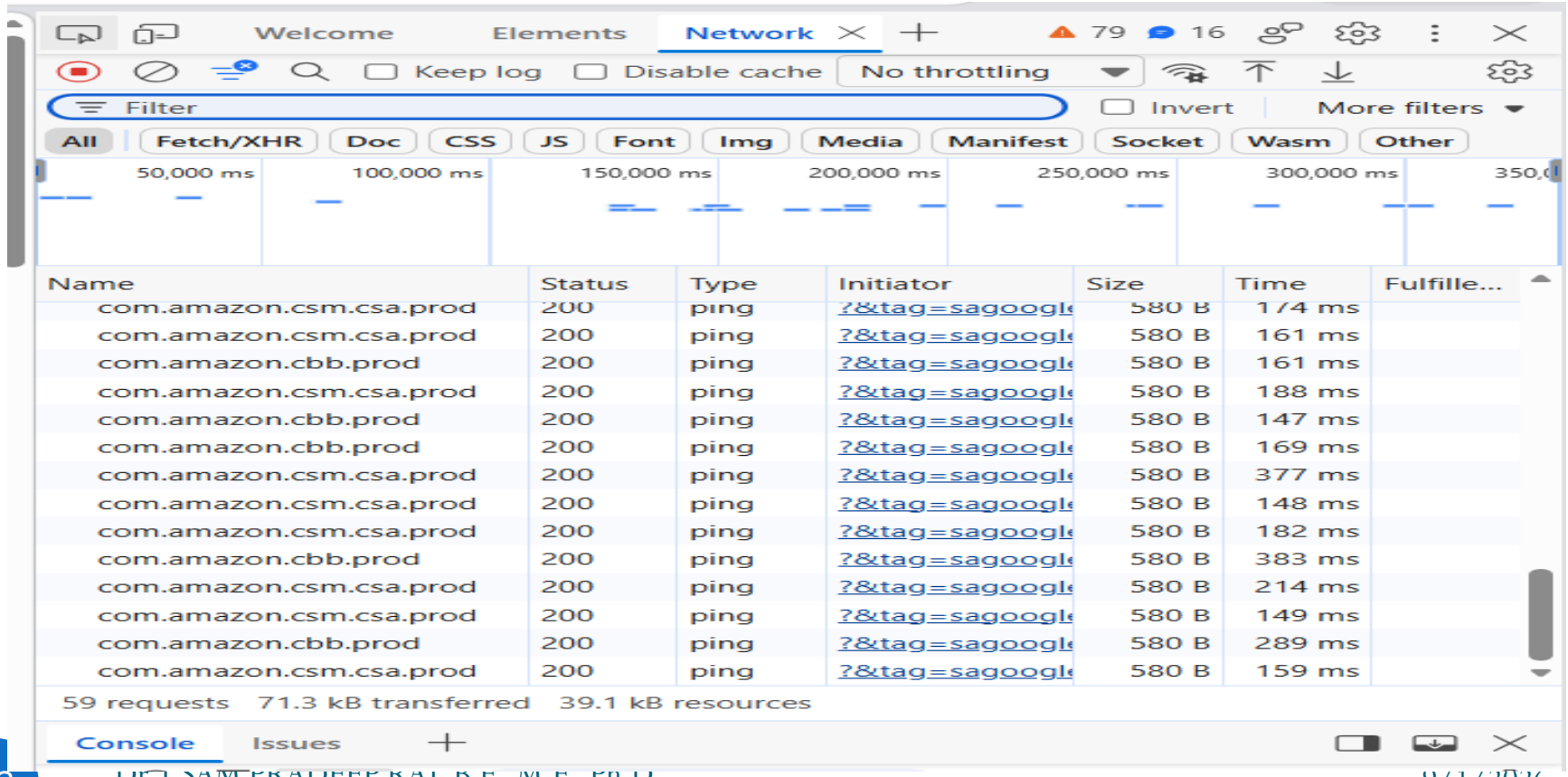
At the top of Developer Tools, click:

**Network**



# HTTP Protocol Demo Using Browser Developer Tools

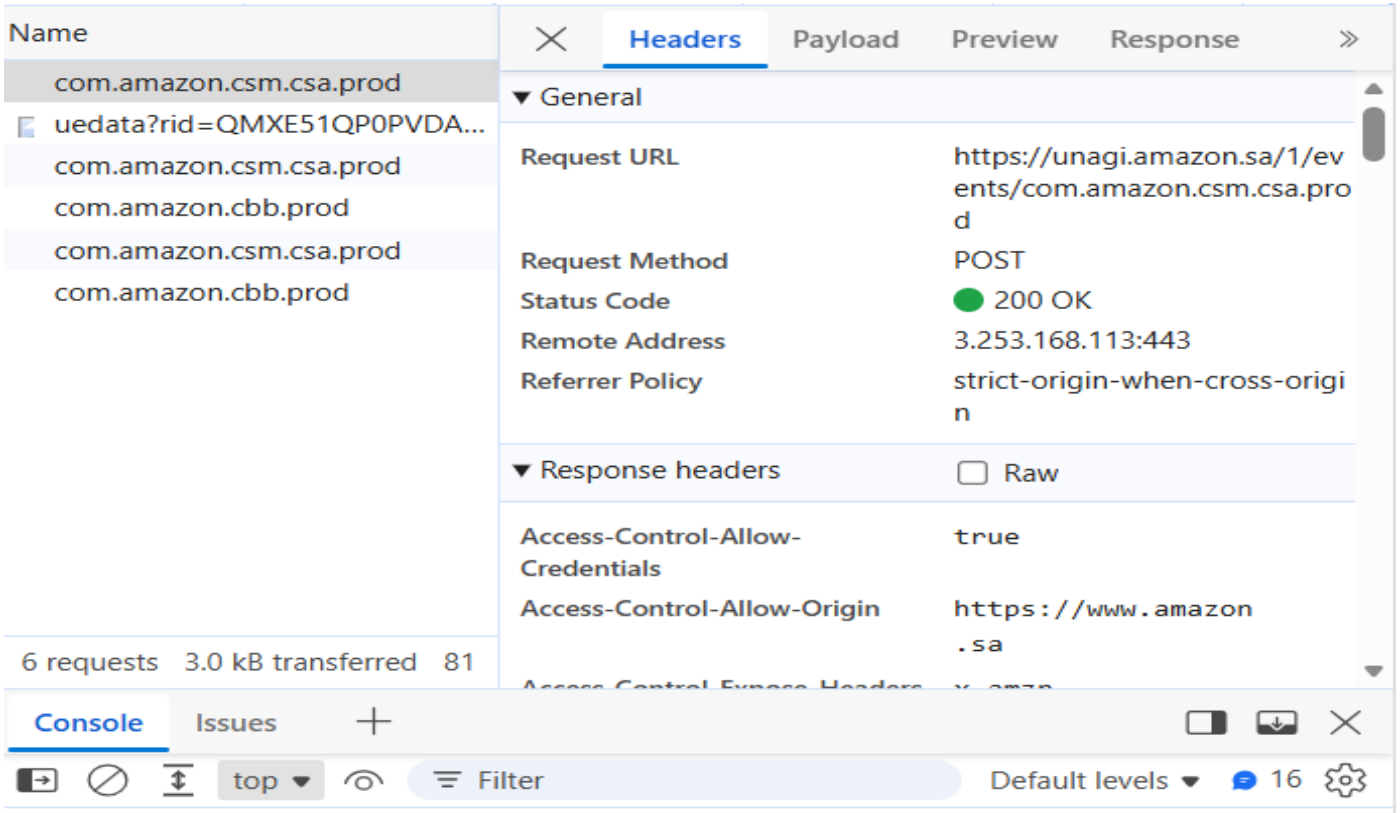
**Step 5:** You may see columns such as: Name, Status ,Type ,Initiator ,Size ,Time



# HTTP Protocol Demo Using Browser Developer Tools

## Step 6: Select a request

Click any row under the Name column.



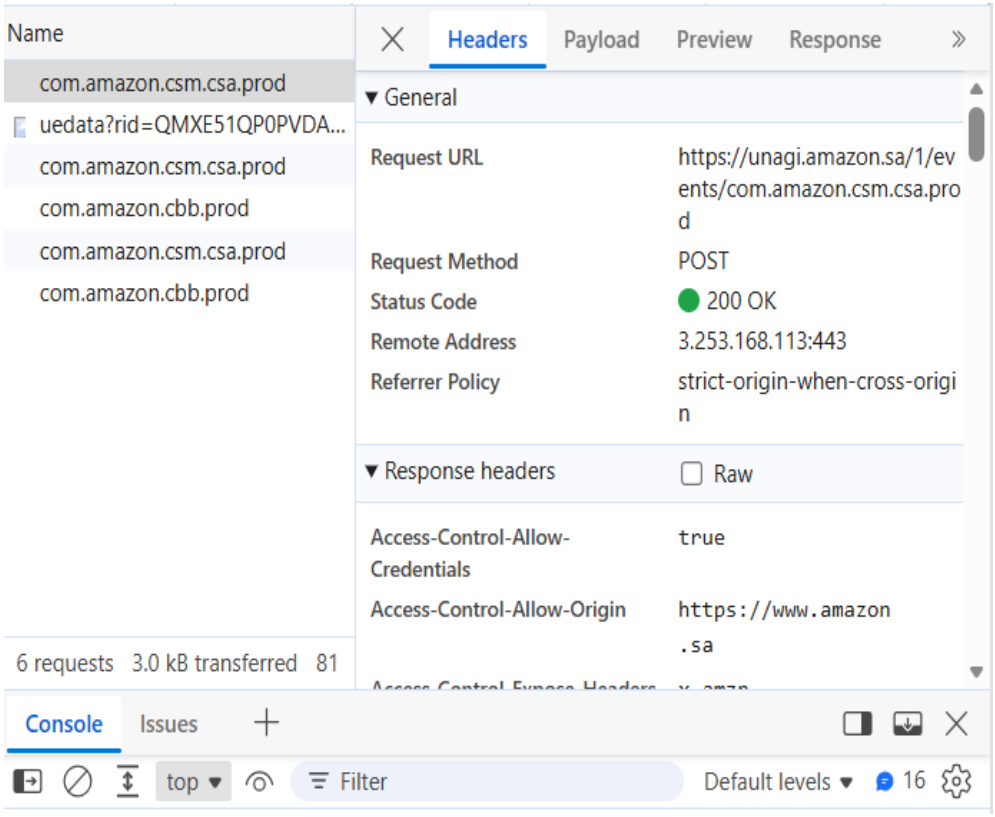
# HTTP Protocol Demo Using Browser Developer Tools

**Step 6: View request details**

After clicking a request, look for the **Headers** section. You can see:

- **Request URL**
- **Request Method** (GET/POST)
- **Status Code**
- **Request Headers**
- **Response Headers**

.



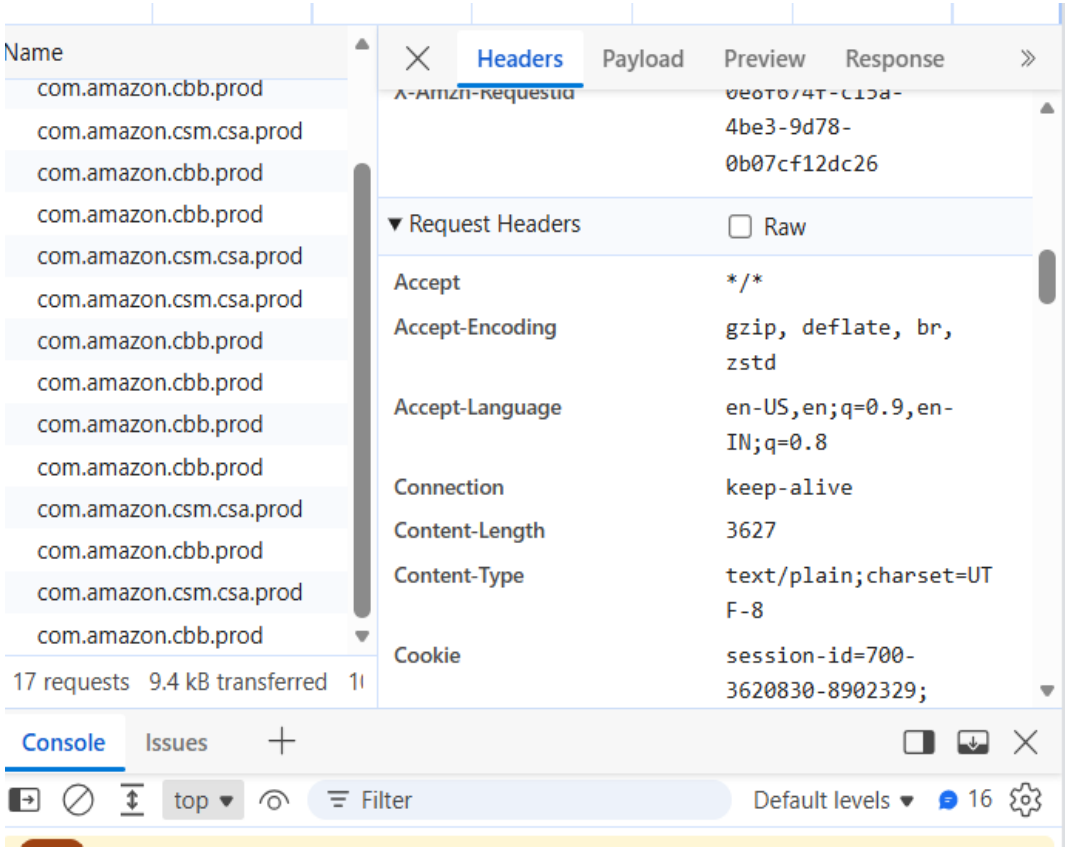
# HTTP Protocol Demo Using Browser Developer Tools

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- **Request Headers**
- **Response Headers**

.



# Learning Outcomes of the HTTP Demonstration

After completing the **Browser Developer Tools → Network tab demonstration**, students can learn the following:

- **How a browser communicates with a web server**
  - The browser sends a request.
  - The server sends a response.
- **HTTP Request Method**
  - Students can observe methods such as **GET** and **POST**.
  - Example: GET is used to request a webpage or resource.

# Learning Outcomes of the HTTP Demonstration

- **HTTP Status Codes**
  - Students can see status codes such as:
  - **200 OK** – Request successful
  - **404 Not Found** – Resource not found
  - **500 Internal Server Error** – Server error
- **Request and Response Headers**
  - Students can observe information exchanged between the browser and server.

# Classroom Activity 1 : MCQ

**1. What is the main purpose of HTTP?**

A) To store files on a computer

B) To transfer web resources between client and server

C) To protect a computer from viruses

D) To create web pages

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**2. Which device/application usually sends an HTTP request?**

A) Web browser
B) Printer
C) Keyboard
D) Monitor

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**3. Which HTTP method is commonly used to request data from a web server?**

A) SEND
B) GET
C) PUSH
D) FETCH

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# Summary: HTTP Protocol

- **HTTP (HyperText Transfer Protocol)** enables communication between a **web browser and a web server**.
- It follows a **Request–Response model**.
- Common HTTP methods include **GET, POST, PUT, and DELETE**.
- HTTP responses use **status codes** such as **200 OK** and **404 Not Found**.
- Browser Developer Tools can be used to **observe requests, responses, headers, and status codes**.
- **HTTPS** is the secure version of HTTP and provides encrypted communication.

# Thank you